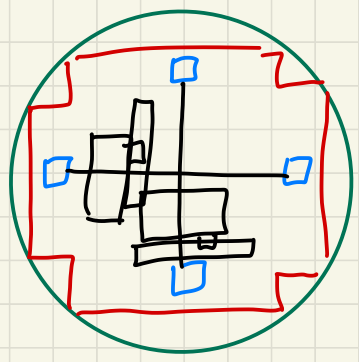
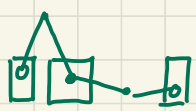
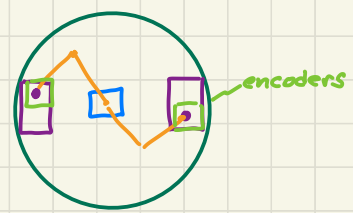
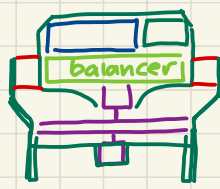
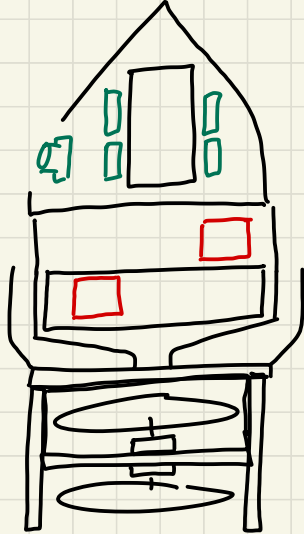
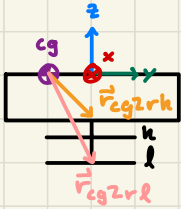




EOM



$$\begin{aligned} H_r &= j_r \omega_r \hat{b}_z \\ \dot{H}_r &= j_r (\dot{\omega}_r \hat{b}_z + \omega_r [\vec{\omega}_b \times \hat{b}_z]) \\ \therefore M_{\text{wheel}} &= -j_r (\dot{\omega}_r \hat{b}_z + \omega_r [\vec{\omega}_b \times \hat{b}_z]) \\ M_{\text{thrust}} &= f(\omega_r) [\vec{r}_{cg2r} \times \hat{b}_z] \end{aligned} \quad \left. \vphantom{\begin{aligned} H_r &= j_r \omega_r \hat{b}_z \\ \dot{H}_r &= j_r (\dot{\omega}_r \hat{b}_z + \omega_r [\vec{\omega}_b \times \hat{b}_z]) \\ \therefore M_{\text{wheel}} &= -j_r (\dot{\omega}_r \hat{b}_z + \omega_r [\vec{\omega}_b \times \hat{b}_z]) \\ M_{\text{thrust}} &= f(\omega_r) [\vec{r}_{cg2r} \times \hat{b}_z] \end{aligned}} \right\}$$

$$\begin{aligned} \vec{r}_{o2mv} &= \vec{r}_{o2mv} \\ \vec{v}_{o2mv} &= \dot{\vec{r}}_{o2mv} + \vec{\omega}_b \times \vec{r}_{o2mv} \\ \vec{a}_{o2mv} &= \ddot{\vec{r}}_{o2mv} + \dot{\vec{\omega}}_b \times \vec{r}_{o2mv} + \vec{\omega}_b \times \dot{\vec{r}}_{o2mv} + \vec{\omega}_b \times [\vec{\omega}_b \times \vec{r}_{o2mv}] \\ \vec{F}_{mv} &= -m_v \vec{a}_{o2mv} \\ \therefore M_{\text{shift}} &= \vec{r}_{cg2mv} \times \vec{F}_{mv} \quad \text{but this is 0 if } r_{z,cg2mv} = 0 \end{aligned}$$

$$\therefore M_{\text{wheel}} + M_{\text{thrust}} + M_{\text{shift}} = \tilde{I}_b \dot{\omega}_b + \omega_b \times (\tilde{I}_b \omega_b)$$

↑
fcn of r_{o2mv} but barely

Linearization:

$$\begin{aligned} &-j_r [\dot{\omega}_{rl} \hat{b}_z + \omega_{rl} [\vec{\omega}_b \times \hat{b}_z] + \dot{\omega}_{rh} \hat{b}_z + \omega_{rh} [\vec{\omega}_b \times \hat{b}_z]] \\ &= -j_r [\hat{b}_z (\dot{\omega}_{rl} + \dot{\omega}_{rh}) + (\vec{\omega}_b \times \hat{b}_z) (\omega_{rl} + \omega_{rh})] \quad M_{\text{wheel}} \end{aligned}$$

if $\omega_{rl} = -\omega_{rh}$, this term = 0 (counter-rotating)

$$\therefore M_{\text{yaw}} \approx -j_r (\dot{\omega}_{rl} + \dot{\omega}_{rh})$$

$$M_{\text{pitch}} \approx r_{cg,x} \cdot K (|\omega_{rl}| + |\omega_{rh}|)$$

$$M_{\text{roll}} \approx r_{cg,y} \cdot K (|\omega_{rl}| + |\omega_{rh}|)$$

$$t \approx K (|\omega_{rl}| + |\omega_{rh}|)$$

$$\dot{x} = \frac{1}{\tau} (-x + u_x)$$

$$\therefore x = \tau \dot{x} - u_x$$

$$\therefore u_x = \tau \dot{x} - x$$

$$\begin{aligned}\dot{\omega}_z &\approx -j_r K (\dot{t}_l - \dot{t}_h) \frac{1}{I_{zz}} \\ \dot{\omega}_y &\approx \Delta x \cdot (t_h + t_l) \frac{1}{I_{yy}} \\ \dot{\omega}_x &\approx \Delta y \cdot (t_h + t_l) \frac{1}{I_{xx}}\end{aligned}$$

$$\begin{aligned}t &= t_h + t_l \\ \dot{t}_h &= \frac{1}{T_r} (-t_h + U_{th}) \\ \dot{t}_l &= \frac{1}{T_r} (-t_l + U_{tl}) \\ \Delta \dot{x} &= \frac{1}{T_\Delta} (-\Delta x + U_{\Delta x}) \\ \Delta \dot{y} &= \frac{1}{T_\Delta} (-\Delta y + U_{\Delta y})\end{aligned}$$

$$\begin{aligned}\dot{x} &= 1/r (-x + U_x) \\ \therefore x &= r \dot{x} - U_x \\ \therefore U_x &= r \dot{x} - x\end{aligned}$$

better yet...

$$\omega_v = \pm k t$$

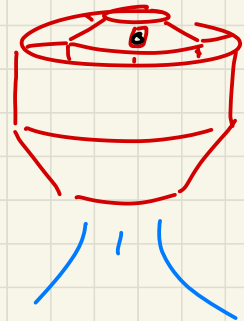
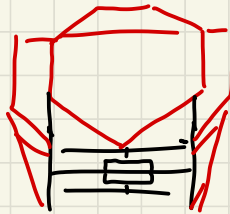
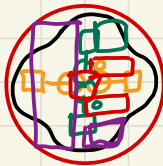
$$\begin{aligned}x &= [\omega_b, \Delta x, \Delta y, t_h, t_l, t] \\ T &= \text{Set point exactly}\end{aligned}$$

$$M_x = \Delta y \cdot t$$

$$\begin{aligned}M_x / t &= \Delta y \\ \uparrow \\ \text{cmd}\end{aligned} \quad \begin{aligned}M_z &= \Delta t, \quad \frac{T}{2} - \Delta t = t_{h, \text{cmd}} \\ \uparrow \\ \text{cmd}\end{aligned} \quad \frac{T}{2} + \Delta t = t_{l, \text{cmd}}$$

$$\begin{aligned}M_y / t &= \Delta x \\ \uparrow \\ \text{cmd}\end{aligned}$$

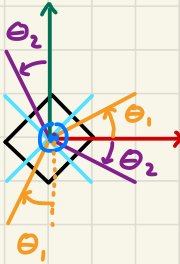
$$\begin{aligned}d_{x, \text{cmd}} &= \text{PID} & \Delta x_{\text{cmd}} &= d_{y, \text{cmd}} / \min(T_{\text{cmd}}, 5\%) \\ d_{y, \text{cmd}} &= \text{PID} & \Delta y_{\text{cmd}} &= d_{x, \text{cmd}} / \min(T_{\text{cmd}}, 5\%) \\ d_{z, \text{cmd}} &= \text{PID} & t_{h, \text{cmd}} &= d_{z, \text{cmd}} \\ T_{\text{cmd}} & & t_{l, \text{cmd}} &= -d_{z, \text{cmd}}\end{aligned}$$



imu $\rightarrow$ transform $\rightarrow$ LPF - $\omega_{b,est}$

$\omega_{b,est} \rightarrow$ PID $\rightarrow d_{cmd}$
 $\omega_{b,des} \rightarrow$

$d_{cmd} \rightarrow$ allocator $\rightarrow act_{cmd}$
 $t_{cmd} \rightarrow$



$$x_{cg} = \cos\theta_1 + \cos\theta_2 - \sin\theta_1 - \sin\theta_2$$
$$y_{cg} = \sin\theta_1 - \sin\theta_2 - \cos\theta_1 + \cos\theta_2$$

$$\theta_1 = \arccos\left(\frac{x_{cg} - y_{cg}}{2\sqrt{2}}\right) - \frac{\pi}{4}$$

$$\theta_2 = \arccos\left(\frac{x_{cg} + y_{cg}}{2\sqrt{2}}\right) - \frac{\pi}{4}$$

$$20 \leq \theta_1 \leq 70$$

$$20 \leq \theta_2 \leq 70$$

$\uparrow$
allocator math

also scaled by throttle cus

$$M = f \cdot d$$

$$\therefore d_{cmd} \rightarrow \frac{d}{f} \rightarrow d_{cmd} \rightarrow alloc \rightarrow \theta_{cmd}$$

will def wanna balance; we can do that
via cowing

